

33. Scatter Plot with Empty Circles

Question

Write a Python program to draw a scatter plot with empty circles taking a random distribution in X and Y and plotted against each other.

Aim

To create a scatter plot using randomly generated X and Y values with empty circle markers.

Code import matplotlib.pyplot as plt import numpy as np

```
# Generate random X and Y values
```

```
np.random.seed(10)
```

```
x = np.random.rand(50) y
```

```
= np.random.rand(50)
```

```
# Scatter plot with empty circles plt.scatter(x, y,
```

```
facecolors='none', edgecolors='blue')
```

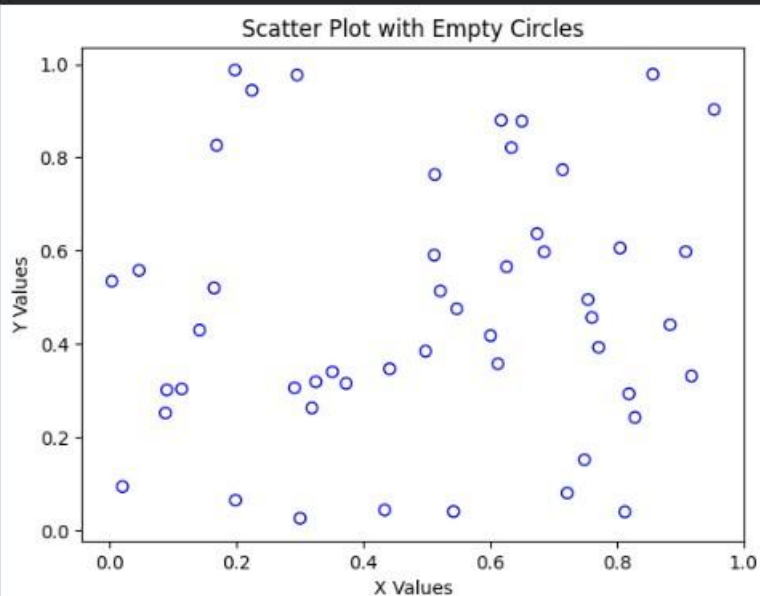
```
# Labels and title plt.xlabel("X Values")
```

```
plt.ylabel("Y Values") plt.title("Scatter
```

```
Plot with Empty Circles")
```

```
plt.show() Output:
```

```
# Labels and title
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.title("Scatter Plot with Empty Circles")
plt.show()
```



34. Scatter Plot with Balls of Different Sizes

Question

Write a Python program to draw a scatter plot using random distributions to generate balls of different sizes.

Aim

To create a scatter plot with randomly distributed points having different sizes.

Code import matplotlib.pyplot as plt import numpy as np

```
# Generate random values
```

```
np.random.seed(10)
```

```
x = np.random.rand(50) y
```

```
= np.random.rand(50)
```

```
# Generate random sizes
```

```
sizes = np.random.randint(20, 500, 50)
```

```
# Create scatter plot plt.scatter(x,  
y, s=sizes, alpha=0.6)  
  
# Labels and title plt.xlabel("X Values")  
plt.ylabel("Y Values") plt.title("Scatter Plot  
with Different Ball Sizes")  
  
plt.show()
```

Output:

